

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0785

Gallagher 100 kV
June 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0785. The Interconnection Customer proposes a 1070.4 MW natural gas combined cycle facility interconnecting at the Gallagher 100 kV substation in the Illinois area. The request was received on April 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 224 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$92,926,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gallagher / Enderlin 100 KV	Marsh Fork / New S Rd 100 KV L	676.2	101	63.5
Bushland / Enderlin 100 KV L	Marsh Fork / New S Rd 100 KV	573.2	117	63.5
New South Rd / Marsh Fork 100 KV L	Gallagher / Enderlin 100 KV	486.3	130	59.5
New S Road Substation / Lullwater 100 KV line	Gallagher / Enderlin 100 KV L	443.5	137	59.5
Ann / Bushland 100 KV	Nixon Road Substation / Bushland 100 KV	387.7	131	50.1
Bushland / Nixon Rd Substation 100 KV L	Bushland / Ann 100 KV L	387.0	142	52.3
Marsh Fork Substation / Grand Coulee Number One 100 KV line	Gallagher / Enderlin 100 KV L	342.9	123	40.5
Lost Canyon / Edge Moor 100 KV	Nixon Road / Bushland 100 KV L	339.4	114	39.0
Ann / Edge Moor 100 KV	Nixon Rd / Bushland Substation 100 KV Line	314.8	129	39.0
Bannock / Grand Coulee Number One 100 KV	Gallagher / Enderlin 100 KV	195.1	112	18.7
San German 6401 / Dayton'S Bluff 100 KV Line	Lost Canyon / Mtn Island 100 KV	175.2	110	8.6
Texon / Mountain Island 100 KV	Dayton'S Bluff / Lost Canyon 100 KV	164.8	202	13.0
Lost Canyon / Mtn Island 100 KV L	Dayton'S Bluff Substation / Lost Canyon 100 KV	161.2	282	18.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dayton'S Bluff / Lost Canyon 100 KV L	Lost Canyon / Mtn Island Substation 100 KV	158.1	308	18.6
Kendallville / Rex Brown Substation 100 KV	Jones Chapel / Dekalb 500 KV	145.0	216	5.3
Nixon Rd / Griffiss Air Base 100 KV line	Bushland / Ann 100 KV line	142.5	143	16.0
Rex Brown / Montibello 100 KV	Jones Chapel / Dekalb 500 KV L	142.2	233	5.3
Bannock Substation / Parkchester 100 KV	Ne S Rd / Marsh Fork Substation 100 KV Line	142.1	127	10.9
Cuba City / Farmersville 100 KV	Dekalb / Jones Chapel 500 KV	142.1	197	5.3
Ligonier / Montibello 100 KV	Dekalb / Jones Chapel 500 KV	140.4	244	5.3
Truman / Lookingglass 230 KV	Beaver / Trego 500 KV	137.9	575	9.9
Mcneil / Slocum Rd 230 KV	Richland / Dekalb 500 KV L	134.8	1,072	25.6
Dekalb Substation / Robison Park 230 KV	Jones Chapel / Dekalb 500 KV	134.2	1,515	34.1
Slocum Rd / Dekalb 230 KV	Richland / Dekalb 500 KV	134.0	1,101	25.6
Richland 500-230 kV Auto	Troy Energy Llc / Richland 500 KV	133.8	1,398	32.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gallagher / Enderlin 100 KV	Rebuild line with higher-capacity conductor	\$8,291,000	100% (\$8,291,000)
Bushland / Enderlin 100 KV L	Rebuild line with higher-capacity conductor	\$4,077,000	100% (\$4,077,000)
New South Rd / Marsh Fork 100 KV L	Rebuild line with higher-capacity conductor	\$3,199,000	95% (\$3,038,000)
New S Road Substation / Lullwater 100 KV line	Rebuild line with higher-capacity conductor	\$1,103,000	95% (\$1,048,000)
Ann / Bushland 100 KV	Rebuild line with higher-capacity conductor	\$8,527,000	97% (\$8,233,000)
Nixon Road / Bushland 100 KV	Rebuild line with higher-capacity conductor	\$13,036,000	100% (\$13,036,000)
Marsh Fork / Grand Coulee Number One 100 KV	Rebuild line with higher-capacity conductor	\$5,284,000	93% (\$4,902,000)
Lost Canyon / Edge Moor 100 KV	Rebuild line with higher-capacity conductor	\$6,714,000	91% (\$6,122,000)
Ann / Edge Moor 100 KV	Rebuild line with higher-capacity conductor	\$3,524,000	91% (\$3,213,000)
Bannock / Grand Coulee Number One 100 KV	Rebuild line with higher-capacity conductor	\$3,382,000	95% (\$3,223,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
San German 6401 / Dayton'S Bluff 100 KV Line	Rebuild line with higher-capacity conductor	\$13,328,000	53% (\$7,010,000)
Texon / Mountain Island 100 KV	Rebuild line with higher-capacity conductor	\$2,294,000	40% (\$924,000)
Lost Canyon / Mtn Island 100 KV L	Rebuild line with higher-capacity conductor	\$8,427,000	41% (\$3,491,000)
Dayton'S Bluff / Lost Canyon 100 KV L	Rebuild line with higher-capacity conductor	\$2,944,000	42% (\$1,230,000)
Rex Brown / Kendallville 100 KV	Rebuild line with higher-capacity conductor	\$3,760,000	13% (\$500,000)
Nixon Rd / Griffiss Air Base 100 KV line	Rebuild line with higher-capacity conductor	\$1,595,000	100% (\$1,595,000)
Montibello / Rex Brown 100 KV	Rebuild line with higher-capacity conductor	\$5,586,000	13% (\$743,000)
Parkchester / Bannock Substation 100 KV	Rebuild line with higher-capacity conductor	\$1,781,000	79% (\$1,405,000)
Cuba City / Farmersville 100 KV	Rebuild line with higher-capacity conductor	\$7,630,000	13% (\$1,014,000)
Ligonier / Montibello 100 KV	Rebuild line with higher-capacity conductor	\$5,590,000	13% (\$743,000)
Truman / Lookingglass 230 KV	Rebuild line with higher-capacity conductor	\$17,595,000	13% (\$2,345,000)
Mcneil / Slocum Rd 230 KV	Rebuild line with higher-capacity conductor	\$10,759,000	13% (\$1,430,000)
Dekalb Substation / Robison Park 230 KV	Rebuild line with higher-capacity conductor	\$29,975,000	13% (\$3,983,000)
Dekalb / Slocum Rd 230 KV	Rebuild line with higher-capacity conductor	\$53,250,000	13% (\$7,078,000)
Richland 500-230 kV Auto	Replace transformer with higher-rated unit	\$31,955,000	13% (\$4,252,000)
	Total allocated to project		\$92,926,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.